

Algebra 1 — Tutorial sheet 5

Transcribed from the Moodle assignment page (no PDF was posted). Opened 25 Sep 2026 Due Fri 2 Oct 2026. No separate HW subset named. Part B added 10 Oct 2026 from a board photo.

Part A (Moodle)

0. If you have not done the HW 4 computations, finish them and submit, especially 11(a) and (c).
1. Look at the example in the Sep 17-3 board image: the map $f(x) = Ax$ has a simple diagonal matrix with respect to a “magic” basis. Finding such bases is a goal for the next few lectures.
2. An operator T on V is a *projection* if $T^2 = T$. Show that if T is a projection then $V = \text{im } T \oplus \ker T$ (internal direct sum). Is this true if V is infinite-dimensional? The converse is false (pointed out by Pradyun): find a counterexample. What extra condition gives an “if and only if”? Why the name “projection”?
3. Artin, Chapter 3, problems 5.1, 5.2, 5.3 (direct sums).
- 4 (SMMC 2023). Let $n \geq 1$ and let A, B, C be n -dimensional subspaces of $\mathbb{R}^{2n}$ with $A \cap B = B \cap C = C \cap A = \{0\}$. Prove there are bases $\{a_i\}$ of A , $\{b_i\}$ of B , $\{c_i\}$ of C such that for each i the vectors a_i, b_i, c_i are linearly dependent.
5. Verify that the examples in the Sep 17-5 board image are fields. Review that every concept so far goes through over a general field.

Part B (board)

Notation: $GL_n(F)$ is the set of invertible $n \times n$ matrices with entries in the field F ; $SL_n(F)$ is the subset of those with determinant 1. “Diagonal” below means diagonalizable. The numbers of problems 1 and 2 were cut off at the board’s edge and are inferred from the order.

1.

- (a) Show $|GL_2(\mathbb{F}_p)| = p(p+1)(p-1)^2$. Calculate $|GL_n(\mathbb{F}_p)|$. What about $|SL_n(\mathbb{F}_p)|$? (Consider $\det: GL_n(F) \rightarrow F^\times$.)
- (b) Find for each $n \leq 3$ the number of subspaces of dimension n in $\mathbb{F}_p^3$.

2.

- (a) For $V = F^3$ define T by $T(e_1) = e_2$, $T(e_2) = e_3$, $T(e_3) = e_1$. Decide if T is diagonal if $F = \mathbb{R}, \mathbb{C}, \mathbb{F}_p$ in turn.
- (b) B is a 3×3 matrix with entries in $\mathbb{C}$. Suppose $\text{tr}(B) = \text{tr}(B^2) = 0$ and $\text{tr}(B^3) = 3d$ for $d \neq 0$. Find $\det(B)$.

3. Let W be a T -invariant subspace of V , where T is diagonal on V . Must $T|_W$ also be diagonal?

4. Let T, S be diagonal operators on V . Show that

$TS = ST \iff$ there is a basis of V with respect to which S and T both have diagonal matrices.

5. Let T be an operator on V/F with

$$p_T(x) = \prod_i (x - \lambda_i)^{m_i} = (x - \mu_1) \cdots (x - \mu_n),$$

so as multisets $\{\mu_1, \dots, \mu_n\} = \{\underbrace{\lambda_1, \dots, \lambda_1}_{m_1}, \dots\}$.

(a) Show $\dim \ker(T - \lambda_j I) \leq m_j$.

(b) Show $p_T(T) = 0$. (Understand the meaning carefully!) This is the Cayley–Hamilton theorem (CH thm). First do it for $V = F^n$, T given by a diagonal matrix. Then do the same for T given by a triangular matrix. This does the job! Why?

[Can you prove CH without assuming $p_T(x)$ factors fully?]